

MARADI, Niger

WMO: 610800

Lat: 13.503N

Lon: 7.127E

Elev: 1240

StdP: 14.05

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.3	57.5	12.2	10.7	83.3	17.2	13.7	80.2	23.0	72.5	21.1	72.7	4.2	80	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	21.1	106.2	69.4	105.0	69.4	103.6	69.2	79.8	92.0	79.1	90.9	78.4	90.0	5.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	147.6	84.6	76.3	143.7	84.0	75.6	140.2	83.6	44.4	92.0	43.7	91.0	43.0	90.2	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	15.8	13.5	51.5	109.5	1.8	1.8	50.2	110.8	49.1	111.9	48.1	112.9	46.8	114.2		
			39.5	83.3	1.2	1.9	38.7	84.6	38.0	85.7	37.3	86.8	36.4	88.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	72.5	77.6	85.7	91.3	92.3	88.4	83.7	80.9	83.6	85.3	79.9	73.5
	DBStd	7.08	4.59	5.62	4.27	2.97	3.16	3.73	3.34	2.77	2.70	2.47	3.60	3.63
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12009	699	772	1105	1238	1310	1153	1044	958	1009	1093	898	730
	CDD65	6535	235	352	640	788	845	703	579	493	559	628	448	265
	CDH74	87494	2644	4385	9368	12644	13310	10103	6978	4908	6503	8151	5584	2916
	CDH80	50168	1149	2346	5917	8570	8935	6020	3284	1832	3076	4599	3171	1269
Wind	WSAvg	5.3	7.1	6.5	5.8	4.8	5.1	5.6	5.3	4.2	3.5	3.8	5.0	7.2
Precipitation	PrecAvg	19.3	0.0	0.0	0.0	0.2	0.7	2.4	5.9	7.1	3.2	0.2	0.0	0.0
	PrecMax	28.7	0.0	0.0	0.2	1.5	2.6	6.1	10.6	13.5	8.0	2.9	0.2	0.2
	PrecMin	10.9	0.0	0.0	0.0	0.0	0.0	0.5	1.3	0.1	0.4	0.0	0.0	0.0
	PrecStd	5.2	0.0	0.0	0.0	0.3	0.6	1.3	2.2	2.5	1.8	0.5	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.0	102.8	105.8	108.7	107.9	104.0	98.3	93.3	98.6	101.1	100.5	95.8
		MCWB	61.5	64.0	66.7	67.4	71.3	75.8	76.6	78.6	75.6	68.8	64.9	62.5
	2%	DB	93.1	99.0	104.2	107.1	106.1	101.8	95.7	91.4	95.9	99.9	98.5	92.9
		MCWB	59.7	62.8	65.4	67.7	72.0	75.7	76.0	78.2	77.1	69.0	64.7	61.1
	5%	DB	89.9	96.4	102.5	105.5	104.6	100.0	93.7	89.7	93.9	98.4	96.6	90.0
		MCWB	58.0	61.5	64.6	68.0	72.6	75.8	76.2	77.6	77.5	69.5	63.4	59.1
	10%	DB	86.5	93.0	100.4	103.8	103.0	98.0	91.7	88.0	91.8	96.6	94.5	87.5
		MCWB	56.8	59.8	63.3	67.5	73.0	75.7	76.1	77.0	77.5	70.1	62.2	57.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	69.8	74.0	77.9	79.9	80.1	79.9	80.1	81.0	79.6	70.8	64.2
		MCDB	92.6	93.2	96.5	98.6	97.3	96.3	91.1	90.2	91.9	92.3	92.1	92.9
	2%	WB	61.1	65.2	70.8	76.2	78.4	78.8	78.7	79.2	79.9	78.3	67.0	62.1
		MCDB	90.6	95.5	93.8	96.8	95.3	94.3	89.6	88.7	90.7	90.8	91.9	90.0
	5%	WB	58.9	62.5	67.5	74.8	77.6	78.0	78.0	78.4	79.1	77.3	64.9	60.4
		MCDB	88.1	93.0	99.1	94.9	94.6	92.9	88.6	87.2	89.6	89.2	92.3	88.2
	10%	WB	57.1	60.4	64.9	73.4	76.7	77.3	77.2	77.8	78.4	76.3	63.4	58.5
		MCDB	85.7	91.9	97.4	93.8	94.1	91.7	87.5	86.0	88.4	88.1	91.1	84.9
Mean Daily Temperature Range	5% DB	MDBR	27.7	27.6	27.8	25.6	21.1	18.0	15.5	14.0	17.1	23.6	28.9	27.9
		MCDBR	31.7	30.8	30.2	27.4	23.3	19.9	17.8	16.0	19.8	26.1	30.5	31.1
		MCWBR	13.5	12.6	12.2	11.3	8.6	5.6	5.3	6.0	6.1	8.7	10.8	12.9
	5% WB	MCDBR	30.9	29.5	27.6	22.3	20.4	18.6	16.1	15.3	17.9	20.7	29.4	30.6
		MCWBR	13.8	13.1	12.9	9.9	7.7	6.1	5.8	6.3	6.6	7.2	10.7	13.1
Clear-Sky Solar Irradiance	taub	0.579	0.640	0.693	0.725	0.708	0.681	0.623	0.595	0.598	0.634	0.552	0.541	
	taud	1.696	1.548	1.458	1.451	1.501	1.584	1.752	1.815	1.792	1.652	1.720	1.758	
	Ebn at Noon	229	222	215	208	208	211	224	234	234	221	234	235	
	Edh at Noon	75	90	100	101	94	85	73	69	71	80	72	69	
All-Sky Solar Radiation	RadAvg	1767	1929	2043	2105	2056	1984	1900	1828	1981	1953	1840	1726	
	RadStd	70	62	90	74	70	70	74	69	51	62	47	68	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page